

GitHub Portfolio

Zaira Guzmán

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Summary

Biotechnologist specializing in Bioinformatics and Computational Genomics with a solid background in biological data analysis and synthetic biology. Expert in Python, R, Unix environments, and omics data processing, with demonstrated experience in complex database cleaning, development of efficient workflows, and evaluation of artificial intelligence models. Passionate about open-source tool development and the application of computational methodologies to drive scientific research and sustainability.

Education

Specialization in Bioinformatics and Computational Genomics

October 2025 – Present

- University of Salamanca, Spain

B.S. in Biotechnology Engineering

Completed in 2023

- Universidad Dr. José Matías Delgado, El Salvador

Professional Experience

Science Specialist at [Meridial Marketplace](#) for [Invisible Technologies](#),

December 2024 – Present

- Side-by-side (SxS) comparative evaluation of generative AI models through prompt engineering and performance analysis.
- Management of operational metrics (AHT) and precision data annotation for language model training.
- Auditing of voice interactions and real-time data retrieval to assess the robustness of LLM models.

Genomics and Bioinformatics Intern at [Genomac Institute](#), Virtual

August 2025 – January 2026

- Design and optimization of therapeutic peptides using Machine Learning, utilizing predictive models (CAMPR3 and Anti-CP) to identify candidates with high antimicrobial activity and low toxicity.
- Discovery of biosynthetic genes and secondary metabolites through genomic and metagenomic analysis and computational prediction tools (antiSMASH, BageL4).
- Execution of virtual screening protocols and molecular docking for the validation of protein-ligand interactions.

Administrative Assistant at Arrocería La Palma, El Salvador

October 2023 – April 2025

- Administrative management, supervision of financial statements, internal control, and audit coordination.
- Client relationship management and document control, ensuring precise tracking of every account and record.
- Production data administration and waste management.

Researcher in Water, Sanitation, and Hygiene (WASH) at [Stockholm International Water Institute SIWI \(Volunteer\)](#), El Salvador

November 2023 – June 2024

- Execution of data-driven territorial diagnostics (WASH) to identify critical gaps in access, sanitation, and hygiene in vulnerable communities.
- Leadership of advocacy strategies and technical communication for government authorities and NGOs, transforming scientific findings into decision-making roadmaps.
- Design of sustainable solutions and evidence-based community resilience programs focused on local empowerment and water resource self-management.

Skills

- **New Technologies and Bioinformatics:** Programming oriented towards data analysis (Python, R, Unix environments). Implementation of cloud-based bioinformatics pipelines applied to proteomics. Machine Learning in multi-omics environments. Management of scientific databases.
- **Soft Skills:** Project management (Agile methodologies), scientific communication, technical report writing, planning, cross-functional teamwork, and professional English.

Additional Training and Conferences

Fundamentals of Data-Driven Precision Medicine for Diabetes

December 2025

- Stanford Department of Medicine

Introduction to Proteomics and its Biomedical Applications

December 2025

- IBSAL, Salamanca

Bioinformatics Course 201

October 2025

- Genomac Institute Inc

II Salvadoran Congress of Biotechnology and Molecular Biology (Speaker)

June 2024

- El Salvador

Introduction to Data Analytics

June 2023

- IBM - Coursera

Languages

- Spanish: Native
- English: B2